TOPIC 9

# Lasso Regression: Explained

The original class slides, with a plain explanation under each one

> **NOTE** **How to read this document**
>
> - Every original slide is kept exactly as it was, with its slide number under it
> - Under each slide (or small set of slides) you get a plain explanation: **what you are looking at** and **what it means**
> - The slides come from a recorded class, so a few look like screenshots of a video: that is normal

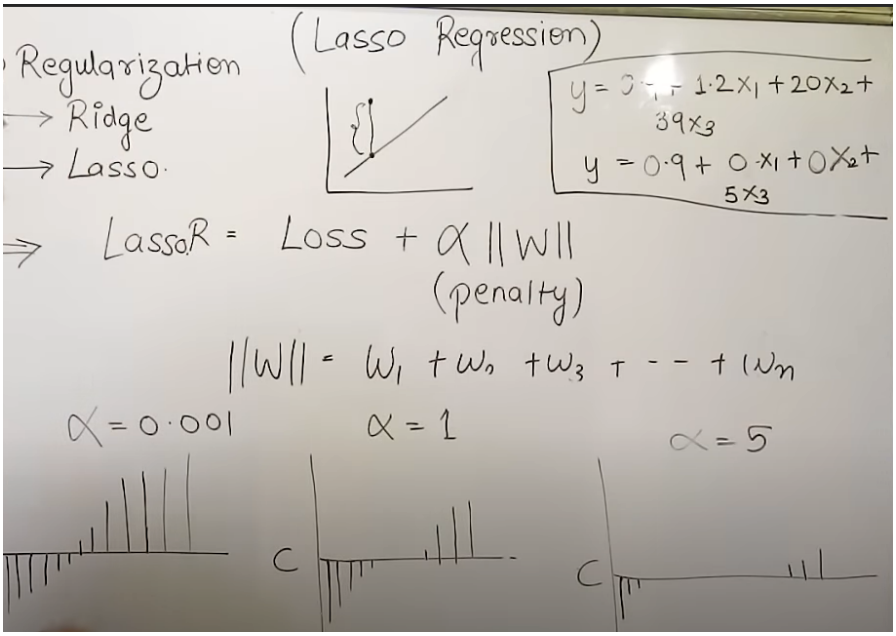
**Lasso regression** is the sibling of Ridge. It uses the **absolute values** of the coefficients as the penalty instead of their squares, and that small change has a big effect: Lasso can push some coefficients to **exactly zero**, which removes those features from the model. These 7 slides show the formula and that zeroing effect.

It is worth reading the Ridge topic first, because the board follows the same layout.

## Words you will meet

| Word | Plain meaning |
|---|---|
| Lasso (L1) | Regularization that uses the sum of absolute values of the coefficients as the penalty |
| ‖w‖ (here) | $\|w_1\| + \|w_2\| + \ldots + \|w_n\|$: the sum of the sizes of the coefficients, ignoring sign |
| α (alpha) | Penalty strength: bigger α means more coefficients pushed to zero |
| Feature selection | Dropping features that do not help; Lasso does this automatically |

## Slide 1: The whole Lasso page



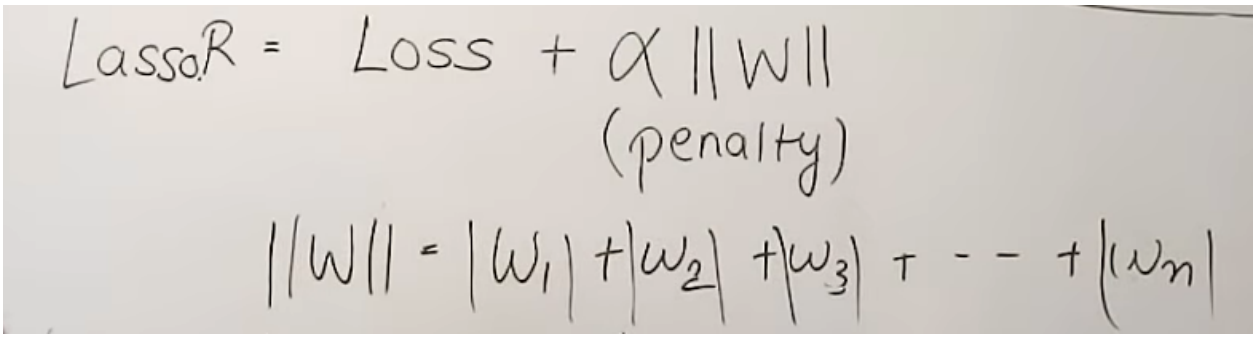

Slide 1

Top left: regularization has two forms, Ridge and Lasso, and Lasso is now being discussed. The formula: **Lasso loss = Loss + α‖w‖**, with ‖w‖ = $|w_1| + |w_2| + \ldots + |w_n|$ as the **penalty**. Top right:

the same two example equations as before. Bottom: bar charts of the coefficients for three strengths: α = 0.001, α = 1 and α = 5.

$$\text{Lasso loss} = \text{Loss} + \alpha(|w_1| + |w_2| + \cdots + |w_n|)$$

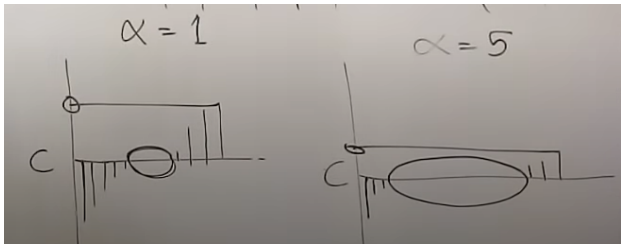
## Slide 2: The Lasso formula up close



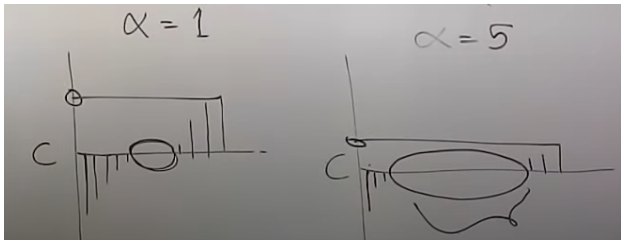

Slide 2

**Lasso loss = Loss + α‖w‖**, where the penalty adds up the **absolute value** of each coefficient: $|w_1| + |w_2| + |w_3| + \ldots + |w_n|$. The vertical bars mean 'ignore the minus sign'. Compare Ridge, which adds up the *squares* ($w_1^2 + w_2^2 + \ldots$). This difference is what makes Lasso set coefficients to exactly zero.

## Slides 3 to 4: Bar charts: α = 1 and α = 5



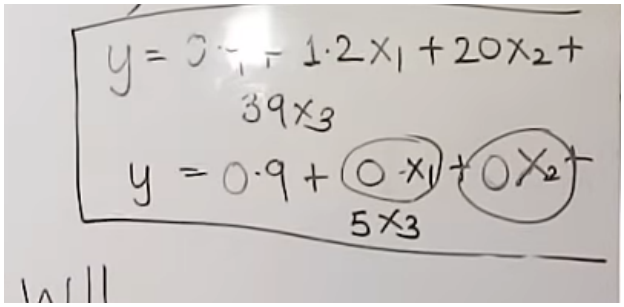

Slide 3



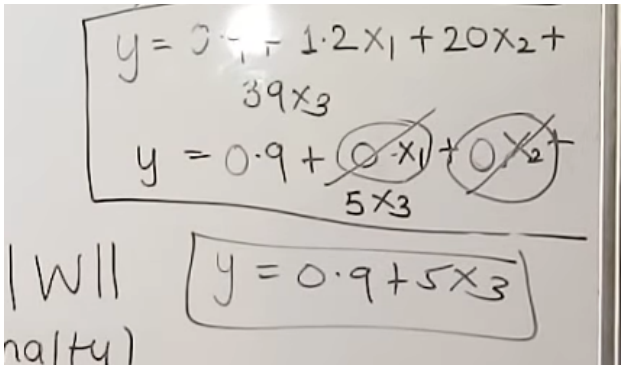

Slide 4

Each bar is one coefficient. At **α = 1** (left) most bars are still there. At **α = 5** (right) the stronger penalty has flattened the whole middle: the circled flat part contains many coefficients that are now **exactly zero** (no bar at all). The few bars that remain at the sides are the features Lasso decided to keep.

## Slides 5 to 6: The equation after Lasso




Slide 5



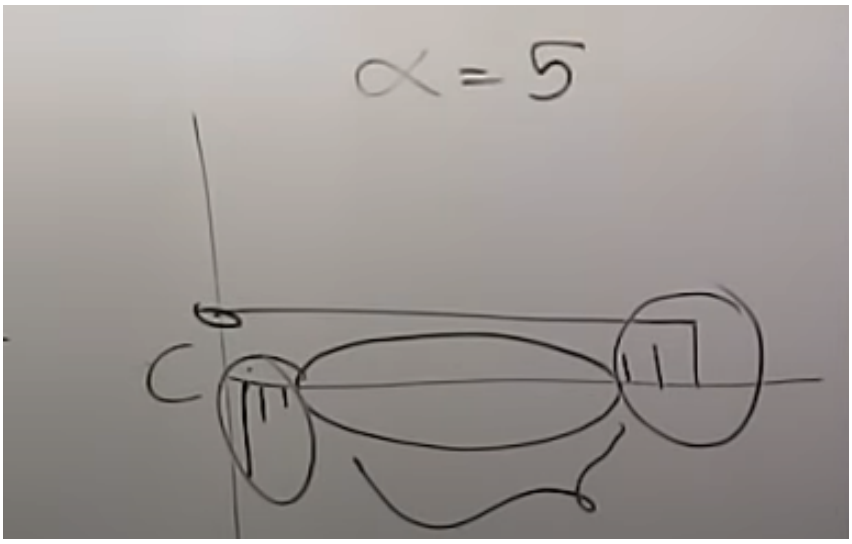

Slide 6

Second equation after Lasso: **y = 0.9 + 0·x₁ + 0·x₂ + 5x₃**. The coefficients of **x₁ and x₂ have become exactly 0**, so those two features are circled and crossed out. They no longer affect the prediction. The model simplifies to **y = 0.9 + 5x₃**.

- Ridge would have made 1.2 and 20 *smaller*

- Lasso makes them *exactly zero* and removes the features

## Slide 7: With a strong penalty, most coefficients vanish




Slide 7

At **α = 5** the left circle and right circle mark the few non-zero coefficients, and the long ellipse in the middle marks the many coefficients that are exactly zero. **A larger α removes more features.**

- Lasso = regularization **plus** automatic feature selection
- Compare with Ridge: Ridge keeps every feature but with smaller coefficients